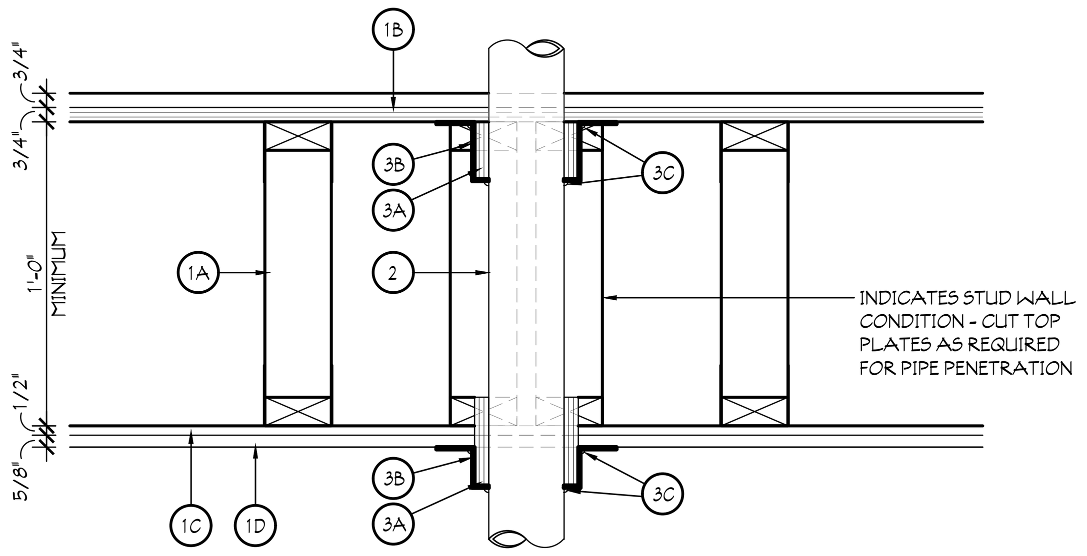
PIPE PENETRATION AT CEILING / FLOOR

(STUD WALLS SIMILAR)

DESIGN NO. FC2020
(FORMERLY SYSTEM NO. 451)
F RATING - 1 HOUR   T RATING - 0 AND 1 HOUR (SEE ITEM 3)



1. FLOOR ASSEMBLY - THE FIRE RATED WOOD TRUSS OR COMBINATION WOOD AND STEEL TRUSS FLOOR - CEILING ASSEMBLY SHALL BE CONSTRUCTED OF THE MATERIALS AND IN THE MANNER DESCRIBED IN THE INDIVIDUAL L500 SERIES DESIGN IN THE UL FIRE RESISTANCE DIRECTORY AND SHALL INCLUDE THE FOLLOWING CONSTRUCTION FEATURES:

A. TRUSSES - MINIMUM 12" DEEP PARALLEL CHORD TRUSSES FABRICATED FROM NOMINAL 2x4 INCH LUMBER IN CONJUNCTION WITH GALVANIZED STEEL TRUSS PLATES OR STRUCTURAL WOOD MEMBERS* WITH BRIDGING AS REQUIRED.

B. FLOORING - NOMINAL 3/4 INCH THICK PLYWOOD FLOORING WITH OR WITHOUT FLOOR TOPPING MIXTURE*, DIAMETER OF HOLE-SAWED OPENING IN FLOOR TO BE A MAXIMUM 1/2 INCH LARGER THAN DIAMETER OF PIPE. MAXIMUM DIAMETER OF OPENING HOLE-SAWED IN FLOORING IS 5 INCHES.

C. FURRING CHANNELS - RIGID OR RESILIENT GALVANIZED STEEL FURRING CHANNELS INSTALLED PERPENDICULAR TO BOTTOM CHORD OF TRUSSES.

D. WALL BOARD, GYPSUM* - NOMINAL 4 FEET WIDE BY 5/8 INCH THICK, SCREW ATTACHED TO FURRING CHANNELS. DIAMETER OF HOLE-SAWED OPENING IN GYPSUM WALLBOARD CEILING TO BE 1/2 INCH MAXIMUM LARGER THAN DIAMETER OF PIPE. MAXIMUM DIAMETER OF OPENING IN CEILING IS 5 INCHES. SECONDARY FIRE STOP SYSTEM (ITEM 3) MUST BE INSTALLED AT UNDERSIDE OF FLOORING PRIOR TO INSTALLATION OF THE GYPSUM WALLBOARD CEILING

2. NONMETALLIC PIPE - NOMINAL 4 INCH DIAMETER (OR SMALLER) SCHEDULE 40 POLYVINYL CHLORIDE (PVC), SDR 17 CHLORINATED POLYVINYL CHLORIDE (CPVC) OR SOLID-CORE SCHEDULE 40 ACRYLONITRILE - BUTADIENE - STYRENE (ABS) PIPE FOR USE IN CLOSED (PROCESS OR SUPPLY) OR VENTED (DRAINS, WASTE, OR VENT) PIPING SYSTEMS. ONE PIPE TO BE INSTALLED APPROXIMATELY MIDWAY BETWEEN TRUSSES AND CENTERED IN CIRCULAR OPENINGS IN FLOOR AND CEILING. PIPE TO BE RIGIDLY SUPPORTED ON BOTH SIDES OF FLOOR - CEILING ASSEMBLY.

3. FIRE STOP SYSTEM - THE HOURLY T RATING FOR FIRE STOP SYSTEM IS DEPENDENT UPON THE TYPE AND SIZE OF NONMETALLIC PIPE, AS TABULATED BELOW:

PIPE TYPE	NOMINAL PIPE DIAMETER, INCHES	ANNULAR SPACE, INCHES	T RATING HOUR
ABS, PVC, OR CPVC	1/2 TO 2	0 TO 1/8	1
ABS	2 1/2 TO 4	0 TO 1/4	0
PVC, OR CPVC	2 1/2 TO 4	0 TO 1/4	1

THE DETAILS OF THE FIRE STOP SYSTEM SHALL BE AS FOLLOWS:

A. FILL VOID OR CAVITY MATERIALS* - WRAP STRIP - NOMINAL 1/4 INCH THICK INTUMESCENT ELASTOMERIC MATERIAL FACED ON ONE SIDE WITH ALUMINUM FOIL, SUPPLIED IN 2 INCH WIDE STRIPS. NOMINAL 2 INCH WIDE STRIP TIGHTLY WRAPPED AROUND NONMETALLIC PIPE (FOIL SIDE EXPOSED) WITH THE EDGES BUTTED AGAINST THE UNDERSIDE OF THE PLYWOOD FLOORING AND AGAINST THE BOTTOM SURFACE OF THE GYPSUM WALLBOARD CEILING AROUND THE ENTIRE PERIMETER OF THE HOLE-SAWED OPENINGS. SUFFICIENT LAYERS OF WRAP STRIP SHALL BE INSTALLED TO LAP A MINIMUM OF 3/16 INCHES ON THE PLYWOOD FLOOR AND ON THE GYPSUM WALLBOARD CEILING AROUND THE ENTIRE PERIMETER OF THE HOLE-SAWED OPENING. FOR NOMINAL 1/2 INCH TO 2 INCH DIAMETER PIPES, A MINIMUM OF ONE LAYER OF WRAP STRIP IS REQUIRED. FOR NOMINAL 2 1/2 INCH AND NOMINAL 3 INCH DIAMETER PIPES, A MINIMUM OF TWO LAYERS OF WRAP STRIP IS REQUIRED. FOR NOMINAL 3 1/2 INCH AND NOMINAL 4 INCH DIAMETER PIPES, A MINIMUM OF THREE LAYERS OF WRAP STRIP IS REQUIRED. EACH LAYER OF WRAP STRIP TO BE INSTALLED WITH BUTTED SEAM, WITH BUTTED SEAMS IN SUCCESSIVE LAYERS STAGGERED. WRAP STRIP LAYER(S) TEMPORARILY HELD IN POSITION USING ALUMINUM FOIL TAPE, STEEL WIRE TIE, OR EQUIVALENT.
MINNESOTA MINING MANUFACTURING COMPANY - TYPE FS-195

B. STEEL COLLAR - NOMINAL 2 INCH DEEP COLLAR WITH 1 1/4 INCH WIDE BY 2 INCH LONG ANCHOR TABS AN D MINIMUM 3/4 INCH LONG TABS TO RETAIN WRAP STRIPS LAYERS. COILS OF PRECUT 0.016 INCH THICK (30 GAUGE) GALVANIZED SHEET STEEL ARE AVAILABLE FROM WRAP STRIP MANUFACTURER. AS AN ALTERNATE, COLLAR MAY BE FIELD FABRICATED FROM MINIMUM 0.016 INCH THICK (30 GAUGE) SHEET STEEL IN ACCORDANCE WITH INSTRUCTION SHEET APPLIED BY WRAP STRIP MANUFACTURER. STEEL COLLAR, WITH ANCHOR TABS BEND OUTWARD 90 DEGREE, WRAPPED TIGHTLY AROUND THE WRAP STRIP LAYERS WITH MINIMUM1 INCH OVERLAP AT THE SEAM. WITH STEEL ANCHOR TABS PRESSED TIGHTLY AGAINST UNDERSIDE OF PLYWOOD FLOORING OR UNDERSIDE OF GYPSUM WALLBOARD CEILING, COMPRESS COLLAR AROUND WRAP STRIP LAYERS USING A MINIMUM 1/2 INCH WIDE BY 0.028 INCH THICK STAINLESS STEEL BAND CLAMP WITH WORM DRIVE TIGHTENING MECHANISM AT THE COLLAR MID HEIGHT. SECURE COLLAR TO PLYWOOD FLOORING USING MINIMUM 3/4 INCH LONG STEEL SCREWS ON CONJUNCTION WITH 1 1/4 INCH DIAMETER STEEL FENDER WASHERS. SECURE COLLAR TO GYPSUM WALLBOARD CEILING USING 3/16 INCH DIAMETER STEEL TOGGLE BOLTS (5/8 INCH GRIP) IN CONJUNCTION WITH 1 1/4 INCH DIAMETER STEEL FENDER WASHERS (MINIMUM OF THREE FASTENERS), SYMMETRICALLY LOCATED, REQUIRED FOR NOMINAL 1/2 INCH TO NOMINAL 3 INCH DIAMETER PIPES. MINIMUM OF FOUR SYMMETRICALLY LOCATED FASTENERS REQUIRED FOR NOMINAL 3 1/2 INCH AND 4 INCH DIAMETER PIPES. AS A FINAL STEP, BEND WRAP STRIP RETAINER TABS 90 DEGREES TOWARD PIPE TO LOCK WRAP STRIP LAYERS IN POSITION.

C. FILL VOID OR CAVITY MATERIALS* - CAULK OR PUTTY - GENEROUS APPLICATION OF CAULK OR PUTTY TO BE APPLIED AROUND THE PERIMETER OF THE STEEL COLLAR AT ITS INTERFACE WITH THE PLYWOOD FLOORING AND GYPSUM WALLBOARD CEILING. A GENEROUS APPLICATION OF CAULK OR PUTTY SHALL BE APPLIED AROUND THE PERIMETER OF THE PIPE AT ITS INTERFACE WITH THE WRAP STRIP LAYERS.
MINNESOTA MINING MANUFACTURER COMPANY - TYPE CP-25 W/B, CP-25 WB+ CAULK, TYPE MP PUTTY

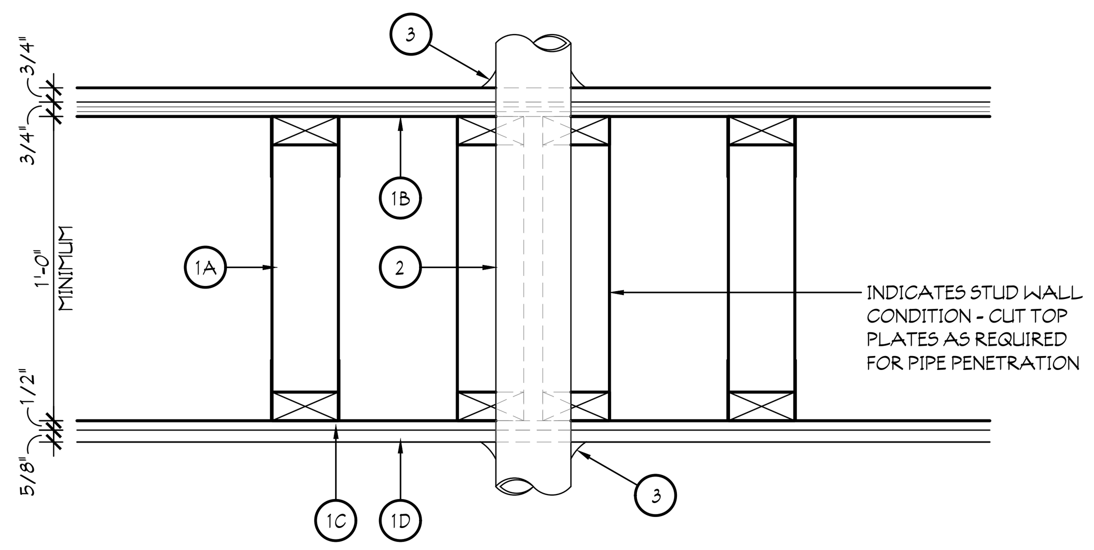
D. FIRE STOP DEVICE* - (NOT SHOWN) - AS AN ALTERNATE TO ITEMS A, B, AND C FOR NOMINAL 1 1/2, 2, 3 OR 4 INCH DIAMETER NONMETALLIC PIPES, A FIRE STOP DEVICE CONSISTING OF A SHEET STEEL SPLIT COLLAR LINE WITH INTUMESCENT MATERIAL AND PROVIDED WITH STEEL CLIPS FOR ATTACHMENT MAY BE USED. FIRE STOP DEVICE TO BE INSTALLED ON UNDERSIDE OF PLYWOOD FLOORING OR UNDERSIDE OF GYPSUM WALLBOARD CEILING AROUND IN ACCORDANCE WITH THE ACCOMPANYING INSTALLATION INSTRUCTIONS.
MINNESOTA MINING AND MANUFACTURING COMPANY - TYPE PPD 150, PPD 200, PPD 300, PPD 400

*BEARING THE UL CLASSIFICATION MARKING

PIPE PENETRATION AT CEILING / FLOOR

(STUD WALLS SIMILAR)

DESIGN NO. FC1006
(FORMERLY SYSTEM NO. 453)
F RATING - 1 HOUR   T RATING - 1 HOUR



1. FLOOR ASSEMBLY - THE FIRE RATED WOOD TRUSS OR COMBINATION WOOD AND STEEL TRUSS FLOOR - CEILING ASSEMBLY SHALL BE CONSTRUCTED OF THE MATERIALS AND IN THE MANNER DESCRIBED IN THE INDIVIDUAL L500 SERIES DESIGN IN THE UL FIRE RESISTANCE DIRECTORY AND SHALL INCLUDE THE FOLLOWING CONSTRUCTION FEATURES:

   A. TRUSSES - MINIMUM 12" DEEP PARALLEL CHORD TRUSSES FABRICATED FROM NOMINAL 2x4 INCH LUMBER IN CONJUNCTION WITH GALVANIZED STEEL TRUSS PLATES OR STRUCTURAL WOOD MEMBERS* WITH BRIDGING AS REQUIRED.

   B. FLOORING - NOMINAL 3/4 INCH THICK PLYWOOD FLOORING WITH OR WITHOUT FLOOR TOPPING MIXTURE*, MAXIMUM DIAMETER OF OPENING HOLE-SAWED IN FLOORING IS 5 INCHES.

   C. FURRING CHANNELS - RIGID OR RESILIENT GALVANIZED STEEL FURRING CHANNELS INSTALLED PERPENDICULAR TO BOTTOM CHORD OF TRUSSES.

   D. WALL BOARD, GYPSUM* - NOMINAL 4 FEET WIDE BY 5/8 INCH THICK, SCREW ATTACHED TO FURRING CHANNELS. MAXIMUM DIAMETER OF HOLE-SAWED OPENING IN GYPSUM WALLBOARD CEILING IS 5 INCHES.

2. PIPE OR CONDUIT - NOMINAL 4 INCH DIAMETER (OR SMALLER) SCHEDULE 10 (OR HEAVIER) STEEL PIPE, STEEL CONDUIT OR STEEL EMT, OR CAST IRON PIPE OR NOMINAL 3 INCH DIAMETER (OR SMALLER) TYPE L (OR HEAVIER) COPPER TUBING. PIPE TO BE INSTALLED APPROXIMATELY MIDWAY BETWEEN TRUSSES AND CENTERED IN CIRCULAR CUTOUTS IN FLOORING (ITEM 1B) AND GYPSUM WALLBOARD CEILING (ITEM 1D). DIAMETER OF CIRCULAR CUTOUTS IN FLOORING AND GYPSUM WALLBOARD CEILING TO BE 1/4 INCH TO 1/2 INCH LARGER THAN DIAMETER OF PIPE. PIPE TO BE RIGIDLY SUPPORTED ON BOTH SIDES OF FLOOR - CEILING ASSEMBLY.

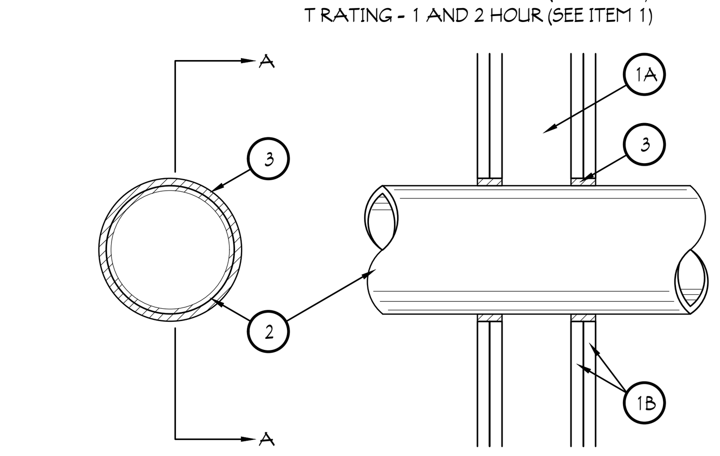
3. FILL VOID OR CAVITY MATERIALS* - CAULK - CAULK FORCED INTO ANNULAR SPACE THROUGHOUT THE THICKNESS OF THE FLOORING AND GYPSUM WALLBOARD CEILING AND WITH A MINIMUM 1/4 INCH DIAMETER OF CAULK APPLIED TO PERIMETER OF PIPE AT ITS EGRESS FROM THE TOP OF THE FLOORING AND THE UNDERSIDE OF THE GYPSUM WALLBOARD CEILING.
MINNESOTA MINING AND MANUFACTURING COMPANY - TYPE CP-25 WB, CP-25 WB+

*BEARING THE UL CLASSIFICATION MARKING

PIPE PENETRATION AT FIRE STOP

(STUD WALLS SIMILAR)

DESIGN NO. WL2067
(FORMERLY SYSTEM NO. 147)
F RATING - 1 AND 2 HOUR (SEE ITEM1)
T RATING - 1 AND 2 HOUR (SEE ITEM 1)



SECTION A-A

1. WALL ASSEMBLY - THE 1 OR 2 HOUR FIRE-RATED GYPSUM WALLBOARD / STUD WALL ASSEMBLY SHALL BE CONSTRUCTED OF THE MATERIALS AND IN THE MANNER SPECIFIED IN THE INDIVIDUAL U300 OR U400 SERIES WALL OR PARTITION DESIGNS IN THE UL FIRE RESISTANCE DIRECTORY AND SHALL INCLUDE THE FOLLOWING CONSTRUCTION FEATURES:

   A. STUDS - WALL FRAMING MAY CONSIST OF EITHER WOOD STUDS OR STEEL CHANNEL STUDS. WOOD STUDS TO CONSIST OF NOMINAL 2x4 LUMBER SPACED 16 INCHES O.C. STEEL STUDS TO BE MINIMUM OF 2 1/2 INCH WIDE AND SPACED A MAXIMUM OF 24 INCHES O.C.
   B. WALLBOARD - GYPSUM* - 5/8 INCH THICK, 4 FOOT WIDE WITH SQUARE OR TAPERED EDGES. THE GYPSUM WALLBOARD TYPE, THICKNESS, NUMBER OF LAYERS, FASTENER TYPE AND SHEET ORIENTATION SHALL BE AS SPECIFIED IN THE INDIVIDUAL U300 OR U400 SERIES DESIGN IN THE UL FIRE RESISTANCE DIRECTORY. MAXIMUM DIAMETER OF OPENING IS 4 3/8 INCHES.

   THE HOURLY F AND T RATINGS OF THE FIRE STOP SYSTEM ARE EQUAL TO THE HOURLY FIRE RATING OF THE WALL ASSEMBLY IN WHICH IT IS INSTALLED.

2. THROUGH PENETRATIONS - ONE NONMETALLIC PIPE OR CONDUIT TO BE CENTERED WITHIN THE FIRE STOP SYSTEM. THE MAXIMUM DIAMETER OF THE THROUGH PENETRANT AND ANNULAR SPACE WITHIN THE FIRE STOP SYSTEM IS DEPENDENT UPON THE TYPE OF FILL MATERIAL (ITEM 3). PIPE OR CONDUIT TO BE RIGIDLY SUPPORTED ON BOTH SIDES OF THE WALL ASSEMBLY. THE FOLLOWING TYPES AND SIZES OF NONMETALLIC PIPES OR CONDUIT MAY BE USED:

   A. POLYVINYL CHLORIDE (PVC) PIPE - NOMINAL 2 INCH DIAMETER (OR SMALLER) SCHEDULE 40 SOLID CORE PVC PIPE FOR USE IN CLOSED (PROCESS OR SUPPLY) PIPING SYSTEMS.
   B. CHLORINATED POLYVINYL CHLORIDE (CPVC) PIPE - NOMINAL 2 INCH DIAMETER (OR SMALLER) SDR 17 CPVC PIPE FOR USE IN CLOSED (PROCESS OR SUPPLY) PIPING SYSTEMS.
   C. RIGID NONMETALLIC CONDUIT+ - NOMINAL 2 INCH DIAMETER (OR SMALLER) SCHEDULE 40 PVC CONDUIT INSTALLED IN ACCORDANCE WITH ARTICLE 347 OF THE NATIONAL ELECTRICAL CODE (NFPA NO. 70).

3. FILL VOID OR CAVITY MATERIAL* - SEALANT - IN 2 HOUR FIRE RATED ASSEMBLIES, MINIMUM 1 1/4 INCH THICKNESS OF FILL MATERIAL APPLIED WITHIN THE ANNULUS, FLUSH WITH BOTH SURFACES OF WALL. IN 1 HOUR FIRE RATED ASSEMBLIES, MINIMUM 5/8 INCH THICKNESS OF FILL MATERIAL APPLIED WITHIN THE ANNULUS, ON BOTH SURFACES OF WALL. ADDITIONAL FILL MATERIAL TO BE INSTALLED SUCH THAT A MINIMUM 5/8 INCH THICK CROWN IS FORMED AROUND THE PENETRATING ITEM AND LAPPING A MINIMUM 1 INCH BEYOND PERIPHERY OF THE OPENING. THE MAXIMUM DIAMETER OF THE THROUGH PENETRANT AND ANNULAR SPACE WITHIN THE FIRE STOP IS DEPENDENT UPON THE TYPE OF FILL MATERIAL AS TABULATED BELOW:

MAXIMUM DIAMETER OF THROUGH PENETRANT	NOMINAL ANNULAR SPACE INCHES	FILL MATERIAL TYPE
1	1/2	EP
2	1	I

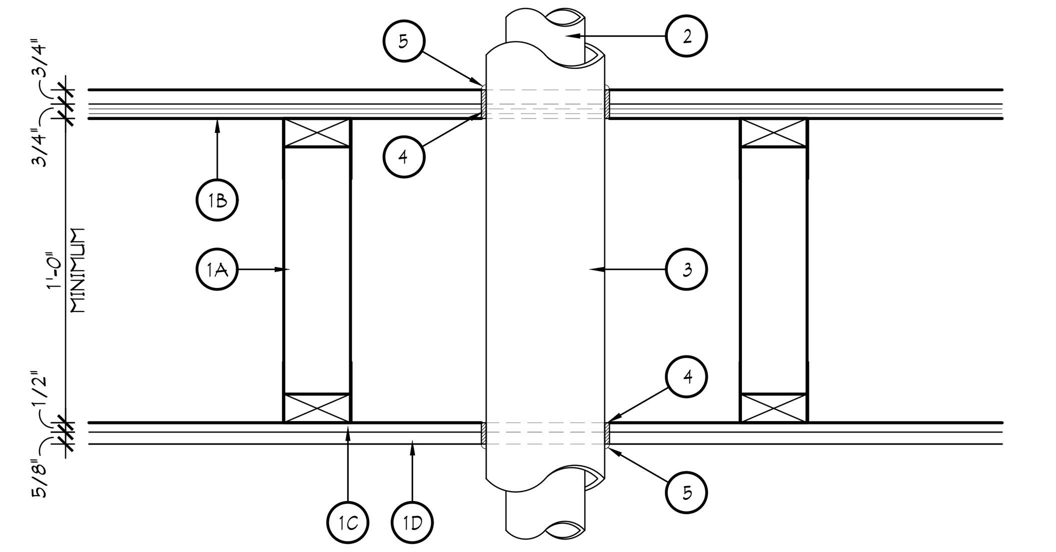
ISOLATEX INTERNATIONAL - TYPES EP AND I

+BEARING THE UL LISTING MARK

*BEARING THE UL CLASSIFICATION MARKING

PIPE PENETRATION AT CEILING / FLOOR

DESIGN NO. FC5002
(FORMERLY SYSTEM NO. 454)
F RATING - 1 HOUR   T RATING - 1 HOUR



1. FLOOR ASSEMBLY - THE FIRE RATED WOOD TRUSS OR COMBINATION WOOD AND STEEL TRUSS FLOOR - CEILING ASSEMBLY SHALL BE CONSTRUCTED OF THE MATERIALS AND IN THE MANNER DESCRIBED IN THE INDIVIDUAL L500 SERIES DESIGN IN THE UL FIRE RESISTANCE DIRECTORY AND SHALL INCLUDE THE FOLLOWING CONSTRUCTION FEATURES:

   A. TRUSSES - MINIMUM 12" DEEP PARALLEL CHORD TRUSSES FABRICATED FROM NOMINAL 2x4 INCH LUMBER IN CONJUNCTION WITH GALVANIZED STEEL TRUSS PLATES OR STRUCTURAL WOOD MEMBERS* WITH BRIDGING AS REQUIRED.

   B. FLOORING - NOMINAL 3/4 INCH THICK PLYWOOD FLOORING WITH OR WITHOUT FLOOR TOPPING MIXTURE*, MAXIMUM DIAMETER OF OPENING HOLE-SAWED IN FLOORING IS 7 INCHES.

   C. FURRING CHANNELS - RIGID OR RESILIENT GALVANIZED STEEL FURRING CHANNELS INSTALLED PERPENDICULAR TO BOTTOM CHORD OF TRUSSES.

   D. WALL BOARD, GYPSUM* - NOMINAL 4 FEET WIDE BY 5/8 INCH THICK, SCREW ATTACHED TO FURRING CHANNELS. MAXIMUM DIAMETER OF HOLE-SAWED OPENING IN GYPSUM WALLBOARD CEILING IS 7 INCHES.

2. PIPE OR CONDUIT - NOMINAL 4 INCH DIAMETER (OR SMALLER) SCHEDULE 10 (OR HEAVIER) STEEL PIPE OR NOMINAL 3 INCH DIAMETER (OR SMALLER) TYPE L (OR HEAVIER) COPPER PIPE. PIPE TO BE INSTALLED APPROXIMATELY MIDWAY BETWEEN TRUSSES AND CENTERED IN CIRCULAR CUTOUTS IN FLOORING (ITEM 1B) AND GYPSUM WALLBOARD CEILING (ITEM 1D). DIAMETER OF CIRCULAR CUTOUTS IN FLOORING AND GYPSUM WALLBOARD CEILING TO BE 1/2 INCHES TO 3/4 INCHES LARGER THAN OUTSIDE DIAMETER OF PIPE COVERING (ITEM 3) OR TUBE INSULATION (ITEM 3A) ON PIPE. PIPE TO BE RIGIDLY SUPPORTED ON BOTH SIDES OF FLOOR - CEILING ASSEMBLY.

3. PIPE COVERING* - NOMINAL 1 INCH THICK HOLLOW CYLINDRICAL HEAVY DENSITY (MINIMUM 3.5 PCF) GLASS FIBER UNITS JACKETED ON THE OUTSIDE WITH AN ALL SERVICE JACKET. LONGITUDINAL JOINTS SEALED WITH METAL FASTENERS OR FACTORY-APPLIED SELF-SEALING LAP TAPE. TRANSVERSE JOINTS SECURED WITH METAL FASTENERS OR WITH BUTT STRIP TAPE SUPPLIED WITH THE PRODUCT. SEE PIPE AND EQUIPMENT COVERING - MATERIALS* (BRGU) CATEGORY IN BUILDING MATERIALS DIRECTORY FOR NAMES OF MANUFACTURERS. ANY PIPE COVERING MATERIAL MEETING THE ABOVE SPECIFICATIONS AND BEARING THE UL CLASSIFICATION MARKING WITH A FLAME SPREAD INDEX OF 25 OR LESS AND A SMOKE DEVELOPED INDEX OF 50 OR LESS MAY BE USED.

3A. TUBE INSULATION - PLASTICS+ - AS AN ALTERNATE TO THE GLASS FIBER PIPE COVERING (ITEM 3), NOMINAL 1/2 INCH OR 5/8 INCH THICK ACRYLONITRILE BUTADIENE / POLYVINYL CHLORIDE (AB/PVC) FLEXIBLE FOAM FURNISHED IN THE FORM OF TUBING WITH SKIN MAY BE USED. SEE PLASTICS (QMFZ2) CATEGORY IN THE RECOGNIZED COMPONENT DIRECTORY FOR NAMES OF MANUFACTURERS. ANY RECOGNIZED COMPONENT TUBE INSULATION MATERIAL MEETING THE ABOVE SPECIFICATIONS AND HAVE A UL 94 FLAMMABILITY CLASSIFICATION OF 94-5Va MAY BE USED.

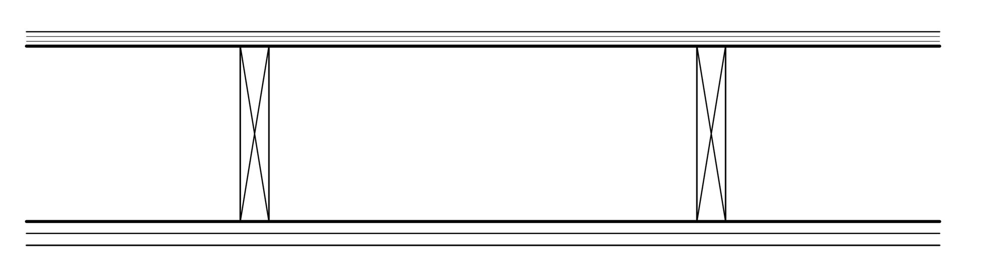
4. FILL VOID OR CAVITY MATERIALS* - WRAP STRIP - NOMINAL 1/4 INCH THICK INTUMESCENT ELASTOMERIC MATERIAL FACED ON ONE SIDE IWTH ALUMINUM FOIL, SUPPLIED IN 2 INCH WIDE STRIPS. NOMINAL 2 IN CH WIDE STRIP TIGHTLY WRAPPED AROUND PIPE COVERING OR TUBE INSULATION (FOIL SIDE EXPOSED), SECURED WITH TWO STEEL TIE WIRES AND SLID INTO HOLE-SAWED OPENING IN FLOORING (ITEM 1B) AND IN GYPSUM WALLBOARD CEILING (ITEM 1D). BOTTOM EDGE OF WRAP STRIP TO PROJECT 9/16 TO 11/16 INCHES BELOW BOTTOM SURFACE OF FLOORING AND BELOW CEILING SURFACE OF GYPSUM WALLBOARD.
MINNESOTA MINING AND MANUFACTURING COMPANY - TYPE FS-195

5. FILL VOID OR CAVITY MATERIALS* - CAULK - NOMINAL 1/4 INCH THICKNESS OF CAULK TO BE APPLIED TO THE EXPOSED EDGE OF THE WRAP STRIP LAYER (TOP OF FLOORING AND BOTTOM OF GYPSUM WALLBOARD CEILING). GENEROUS APPLICATION OF CAULK TO BE APPLIED TO FILL ALL GAPS AT THE WRAP STRIP / FLOORING AND WRAP STRIP / GYPSUM WALLBOARD CEILING INTERFACES.
MINNESOTA MINING AND MANUFACTURING COMPANY - TYPE CP-25 WB, IC-15 WB+

*BEARING THE UL CLASSIFICATION MARKING

CEILING / FLOOR ASSEMBLY

DESIGN NO. L598 (SIMILAR)
STC 67 MIN.



APPROXIMATE CEILING WEIGHT: 5 PSF
FIRE TEST: FM FC 172, 2-25-72

GYPSUM ASSOCIATION

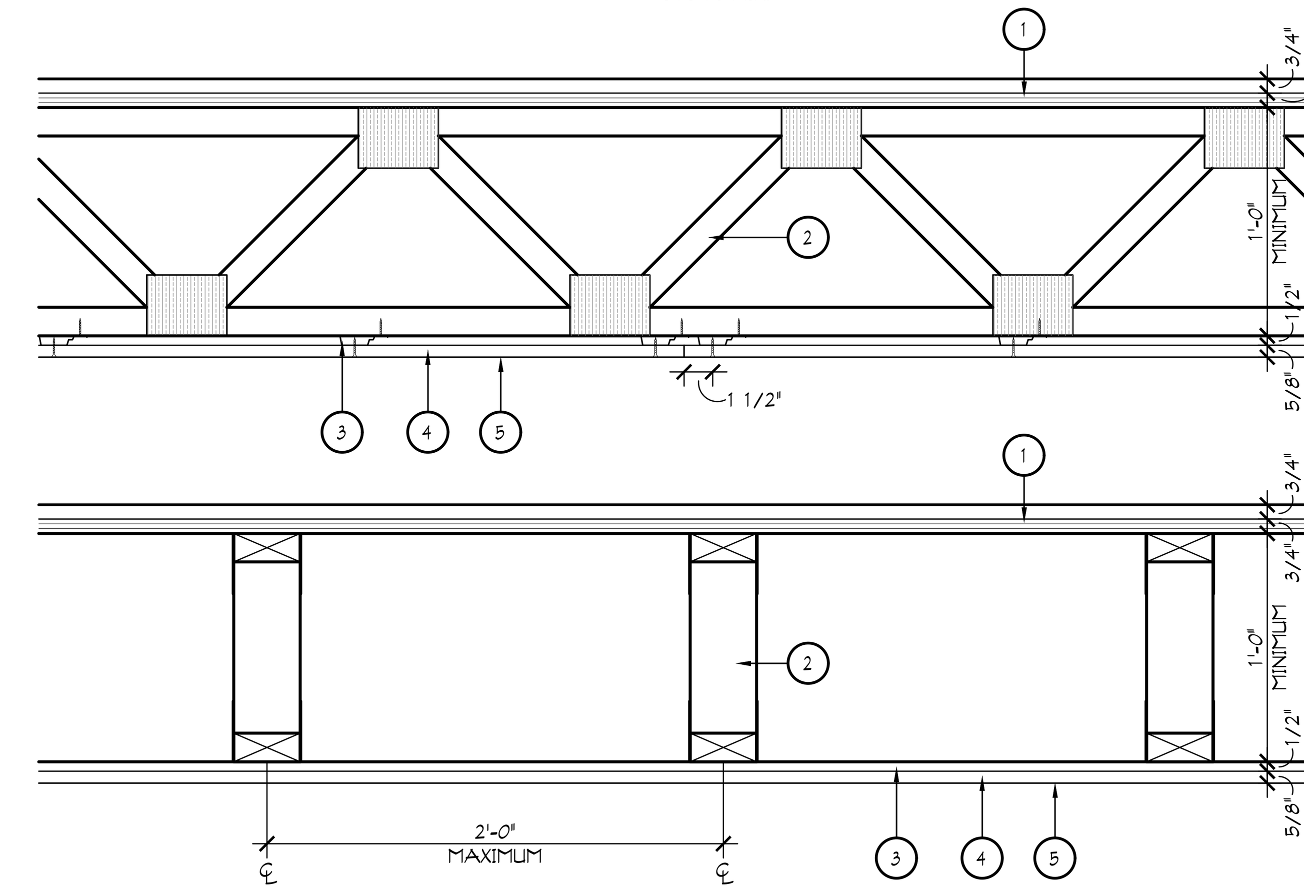
CONSTRUCTION TYPE: WOOD JOISTS, GYPSUM WALLBOARD - BASE LAYER OF 5/8 INCH TYPE "X" GYPSUM WALLBOARD APPLIED AT RIGHT ANGLES TO WOOD JOISTS 24 INCHES O.C. WITH 1 1/4 INCH TYPE S DRWALL SCREWS AT 24 INCHES O.C. FACE LAYER 5/8 INCH TYPE "X" GYPSUM WALLBOARD OR VENEER BASE APPLIED AT RIGHT ANGLES TO JOISTS THROUGH BASE LAYER WITH 1 7/8 INCH TYPE S DRYWALL SCREWS AT 12 INCHES O.C. AT JOINTS AND INTERMEDIATE JOISTS. FACE LAYER JOINTS OFFSET 24 INCHES FROM BASE LAYER JOINTS, 1 1/2 INCH TYPE "G" DRYWALL SCREWS PLACED 2 INCHES BACK ON EITHER SIDE OF FACE LAYER END JOINTS, 12 INCHES O.C. 1/2 INCH PLYWOOD WITH EXTERIOR GLUE APPLIED AT RIGHT ANGLES TO TOP OF JOISTS WITH 8d NAILS. CEILING PROVIDES ONE HOUR FIRE RESISTANCE PROTECTION FOR WOOD FRAMING.

*BEARING THE UL CLASSIFICATION MARKING

- ONE HOUR WALLBOARD FLOOR - CEILING OR ROOF - CEILING ASSEMBLY AND WOOD FRAMING - THE CEILING CONSTRUCTION IS IDENTICAL TO ASSEMBLE FC5406 IN THE GYPSUM ASSOCIATION FIRE RESISTANCE DESIGN MANUAL. ALTERNATE FRAMING MAY CONSIST OF LOWER CHORDS OF TRUSSED RAFTERS, OR PITCHED OR FLAT TRUSSES SPACED A MAXIMUM OF 24 INCHES ON CENTER. ALTERNATE FASTENERS HAVING THE SAME SPACING AS THE SCREWS MAY BE 1 7/8 INCHES 6-PENNY COOLER, BOX OR WALLBOARD NAILS FOR THE BASE PLY AND 2 3/8 INCH 8-PENNY COOLER, BOX OR WALLBOARD NAILS FOR THE FACE PLY. TYPE "G" DRYWALL SCREWS, 1 1/2 INCH LONG ARE STILL REQUIRED AT THE END JOINTS OF THE GYPSUM WALLBOARD FACE LAYER.

CEILING / FLOOR ASSEMBLY

DESIGN NO. L528
UNRESTRAINED ASSEMBLY RATING - 1 HOUR
FINISH RATING - 22 MINUTES
STC 70 MIN.



SYSTEM NO. 5

1. FINISH FLOORING - FLOOR TOPPING MIXTURE* - FROM CONCENTRATE MIXED 40:1 BY VOLUME OF WATER AND EXPANDED AT 100 PSI THROUGH NOZZLE. MIX AT A RATE OF 1.4 CUBIC FEET OF PREFORMED FOAM TO 94 POUNDS OF TYPE I PORTLAND CEMENT AND 300 POUNDS OF SAND WITH 5 1/2 GALLONS OF WATER. CAST DENSITY OF FLOOR TOPPING MIXTURE SHALL BE 100 (+ OR -) 5 PCF WITH A MINIMUM COMPRESSIVE STRENGTH OF 1000 PSI. MINIMUM THICKNESS SHALL BE 3/4"
ELASTIZELL CORPORATION OF AMERICA - TYPE FF

   VAPOR BARRIER - (OPTIONAL) - COMMERCIAL ASPHALT SATURATED FELT, 0.030 INCHES THICK.

   SUB-FLOORING - 23/32 INCH THICK PLYWOOD MINIMUM GRADE "C-D" EXPOSURE 1 OR "C-D" WITH EXTERIOR GLUE CONFORMING TO PS 1-83 SPECIFICATIONS OR NON-VENEER APA RATED SHEATHING PANELS PER APA SPECIFICATIONS PRP-108. LONG DIMENSION OF PANELS (STRENGTH AXIS) OR FACE GRAIN OF PLYWOOD TO BE PERPENDICULAR TO JOIST WITH JOINTS STAGGERED.

2. TRUSSES - PARALLEL CHORD TRUSSES SPACED A MAXIMUM OF 24 INCHES O.C. FABRICATED FROM NOMINAL 2x4 LUMBER WITH LUMBER ORIENTED EITHER VERTICALLY OR HORIZONTALLY. TRUSS MEMBERS SECURED TOGETHER WITH NO. 20 MSG GALVANIZED STEEL TRUSS PLATES. PLATES TO HAVE 5/16 INCH LONG TEETH PROJECTING PERPENDICULAR TO THE PLANE OF THE PLATE. THE TEETH ARE IN PAIRS FACING EACH OTHER (MADE BY THE SAME PUNCH), FORMING A SPLIT-TOOTH-TYPE PLATE. EACH TOOTH HAS A CHISEL POINT ON ITS OUTSIDE EDGE. THESE POINTS ARE DIAGONALLY OPPOSITE EACH OTHER FOR EACH PAIR. THE TOP HALF OF EACH TOOTH HAS A TWIST FOR STIFFNESS. THE PAIR ARE REPEATED ON APPROXIMATE 7/8 INCH CENTER WITH FOUR ROWS OF TEETH PER INCH OF PLATE WIDTH.

3A. FURRING CHANNEL - RESILIENT, FORMED OF 26 MSG GALVANIZED STEEL AS SHOWN, SPACED 16 INCHES O.C. PERPENDICULAR TO TRUSSES. CHANNELS SECURED TO EACH TRUSS WITH TYPE S, 1 1/4 INCH LONG STEEL SCREWS (SPACED 24 INCHES O.C.). CHANNELS OVERLAPPED AT SPLICES BY 4 INCHES.

4. WALLBOARD* - GYPSUM - 5/8 INCH THICK, 4 FOOT WIDE, SHEETS OF WALLBOARD INSTALLED WITH LONG DIMENSION PERPENDICULAR TO FURRING OR RESILIENT CHANNELS WITH 1 INCH LONG WALLBOARD SCREWS SPACED 12 INCHES O.C. AND LOCATED A MINIMUM 1 1/2 INCHES FROM SIDE AND END JOINTS. AT END JOINTS, TWO FURRING OR RESILIENT CHANNELS ARE USED WHICH EXTEND A MINIMUM OF 6 INCHES BEYOND BOTH ENDS OF THE JOINT.
CANADIAN GYPSUM COMPANY,LTD. - TYPE C
CELOTX CORPORATION - TYPE FRP
DOMTAR GYPSUM - TYPE 5
GEORGIA PACIFIC CORPORATION., GYPSUM DIVISION - TYPE GPFS-C
GOLD BOND BUILDING PRODUCTS. - TYPE FSW-G
PABCO GYPSUM COMPANY - TYPE C
AMERICAN GYPSUM COMPANY - TYPE AG-C
UNITED STATES GYPSUM COMPANY - TYPE C OR IP-X2

5. SCREWS, WALLBOARD - 1 INCH LONG, TYPE S, 9/64 INCH DIAMETER, SELF-DRILLING AND SELF-TAPPING BUGLE HEAD

6. FINISHING SYSTEM - (NOT SHOWN) - PAPER TAPE EMBEDDED IN CEMENTITIOUS COMPOUND OVER JOINTS WITH EDGES OF COMPOUND FEATHERED OUT AND EXPOSED SCREW HEADS COVERED WITH COMPOUND. AS AN ALTERNATED, NOMINAL 3/32 INCH THICK VENEER PLASTER MAY BE APPLIED TO THE ENTIRE SURFACE OF GYPSUM WALLBOARD.

*BEARING THE U.L. CLASSIFICATION MARKING

GENERAL NOTES

-ALTHOUGH DRAWINGS ARE DRAWN TO SCALE, THE ACCURACY OF THE DRAWING CANNOT BE GUARANTEED DUE TO DISCREPANCIES IN THE METHODS OF REPRODUCTION. DIMENSIONS SHOULD BE DERIVED FROM DIMENSIONED PLANS.

-ALL DIMENSIONS, ROUGH OPENING SIZES, AND ELEVATION MARKERS MUST BE FIELD VERIFIED BY THE GENERAL CONTRACTOR. MAUST ARCHITECTURAL SERVICES SHALL NOT BE HELD RESPONSIBLE FOR ANY DIMENSIONAL DISCREPANCIES WHICH HAVE NOT BEEN BROUGHT TO THEIR ATTENTION PRIOR TO CONSTRUCTION.

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HOME2 SUITES BY HILTON

SHEET SIZE 24x36

FT. WAYNE, INDIANA

REVISIONS:

PRELIMINARY - NOT FOR CONSTRUCTION

DATE: 05/01/26
JOB NO.: MAS-26-105
NAME: FLOOR AND PIPE PENETRATIONS DETAILS
SHEET: A453